

Longitudinal evidence for the emergence of multiple intelligences in assistance dog puppies

Hannah Salomons^{a,*}, Morgan Ferrans^b, Candler Cusato^a, Kara Moore^c,
Vanessa Woods^a, Emily Bray^d, Brenda Kennedy^e, Theadora Block^e,
Laura Douglas^e, Ashton Roberts^e, Margaret Gruen^c, Brian Hare^{a,b,f}

^a Department of Evolutionary Anthropology, Duke University, Durham, NC, U.S.A.

^b Center for Cognitive Neuroscience, Duke University, Durham, NC, U.S.A.

^c Department of Clinical Sciences, College of Veterinary Medicine, North Carolina State University, Raleigh, NC, U.S.A.

^d College of Veterinary Medicine, University of Arizona, Tucson, AZ, U.S.A.

^e Canine Companions, Santa Rosa, CA, U.S.A.

^f Psychology and Neuroscience Department, Duke University, Durham, NC, U.S.A.

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Cognitive battery testing suggests that adult domestic dogs, *Canis familiaris*, have different types of cognitive abilities that vary independently. In the current study, we examined whether these cognitive skills develop and emerge independently in puppies by testing them over a crucial period of development. Service dog puppies ($N = 113$) were raised using two different socialization strategies and were either tested longitudinally ($N = 91$) or at a single time point ($N = 22$). Subjects tested longitudinally participated in the battery every 2 weeks, from approximately 8–20 weeks of age, during and just beyond their final period of rapid brain growth. Control puppies only participated in the test battery once, which allowed us to evaluate the impact of repeated testing. In support of the multiple intelligences hypothesis (MIH), cognitive skills emerged at different points across development, not simultaneously. Maturation patterns also varied between cognitive skills, with puppies showing adult-like performance on some tasks only weeks after a skill emerged, while never achieving adult performance in others. Differences in rearing strategy did not lead to differences in developmental patterns while, in some cases, repeated testing did. Overall, our findings provide strong support for the MIH by demonstrating differentiated development across the cognitive abilities tested.

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Domestic dogs, *Canis familiaris*, have a range of cognitive skills they use to flexibly solve problems in the physical and social world (Bräuer & Kaminski, 2021; Hare & Ferrans, 2021; Hare & Woods, 2013; Kaminski & Marshall-Pescini, 2014; Miklósi, 2014). Cognitive test batteries provide evidence that dogs have different types of cognitive abilities that vary independently. When large numbers of dogs ($N > 500$) have been tested across a range of different problems, individual skills for solving related cognitive tasks are often correlated. However, performance on tasks designed to recruit different forms of cognition (e.g. physical reasoning skills versus social skills) have remained largely uncorrelated (Bray et al., 2014; Gnanadesikan et al., 2020; Horschler et al., 2019; MacLean,

Herrmann, et al., 2017; Stewart et al., 2015). Performance within and across individuals was also stable on several cognitive measures when tested longitudinally as part of a larger test battery. Puppies who were above or below average performance at 10 weeks tended to maintain their position on a subset of tasks relative to other dogs when tested again as 2-year-old adults (Bray, Gruen, et al., 2021).

These individual differences are relevant in the real world. Cognitive performance on a range of social and nonsocial tasks is associated with and predictive of training performance in working dogs (Bray et al., 2019; Bray, Otto, et al., 2021; Bray, Sammel, Cheney, et al., 2017; Hare & Ferrans, 2021; Lazarowski et al., 2019, 2020; MacLean & Hare, 2018). Cognitive differences across individuals have also been linked to training success and unwanted behaviours in pet dogs (Junttila et al., 2024). These same individual differences may be targeted with selection since they

* Corresponding author.

E-mail address: hannah.salomons@duke.edu (H. Salomons).

are heritable (with cognitive abilities previously identified as independent also varying in their degree of heritability) (Bray, Gnanadesikan, et al., 2021; Gnanadesikan et al., 2020).

This body of work provides strong support for the hypothesis that adult dogs have multiple types of intelligence (Hare & Ferrans, 2021), which we will refer to as the multiple intelligences hypothesis (MIH). The MIH stems from the idea that evolution shapes organisms to solve the problems they are most likely to face in their environments in ways that promote their survival and reproduction (Hare, 2001; Hare & Wrangham, 2002; Tomasello & Call, 1997). Different cognitive abilities provide different forms of flexible problem solving to overcome a range of novel challenges that an organism predictably encounters (e.g. finding food, competing for mates, cooperating; Shettleworth, 2009; Tomasello, 2022). The MIH does not preclude a hierarchical structure in which variation in different cognitive skills is, in part, explained by a higher-order cognitive variable(s) (i.e. Bognár et al., 2024; Herrmann et al., 2010; Mithen, 1996; Sternberg, 1999). However, it does stand in stark contrast to any hypothesis positing that a single, general form of intelligence or unidimensional cognitive factor such as 'learning ability' can explain cognitive diversity across individuals and species (Bastos & Taylor, 2020; Deaner et al., 2007; Macphail & Bolhuis, 2001; Shettleworth, 2009; Tomasello & Call, 1997). We will call this alternative the general intelligence hypothesis (GIH).

Cognitive ontogeny provides a critical test of these two models of cognitive architecture. The MIH predicts cognitive abilities should show differentiated developmental patterns (Herrmann et al., 2010; Wobber et al., 2014). As the brain develops, the different forms of cognition it supports should emerge separately. The timing of emergence and maturation of cognitive abilities are predicted to differ, in part, because brain areas that support different types of cognition can develop at different rates. Alternatively, the GIH predicts that cognitive abilities should show a unified developmental pattern. Emergence and development could vary across individuals, based largely on experiential differences, but no clear order of cognitive emergence should be observed across a group between different cognitive tasks, and no maturational pace should be specific to a certain cognitive task. Therefore, finding differentiated development patterns would provide a powerful falsification of the GIH.

Developmental comparisons provide preliminary support for the MIH. Compared to wolf puppies, domestic dog puppies show early emerging use of human gestures and eye contact but similar memory and self-control abilities (Salomons et al., 2021). Lazarowski et al. (2020) tested a group of dogs raised, bred and trained for detection work using a cross-sectional design, and found that dogs were skilled by 6 months in a self-control task while skill with a delayed memory task did not develop until 11 months. While these previous studies support the developmental prediction of the MIH, they have relatively low resolution since they only tested any given puppy at a single time point during their early development.

By 8 weeks a puppy is showing some key features of independence, such as being weaned and walking, but its brain is still rapidly developing until at least 16 weeks of age (Gross et al., 2010). Dog brains, like those of humans, rely on white matter to rapidly communicate signals between areas of the brain. The relative white to grey matter intensity and myelination of critical neuron networks begins approaching adult levels by 16 weeks (Gross et al., 2010). The full length of the corpus collosum, the part of the brain that connects the two hemispheres and allows them to communicate, also reaches adult form around 16 weeks of age (Gross et al., 2010; Schmidt et al., 2012). While a puppy's brain

continues to develop after 16 weeks, the period between weaning and 16 weeks represents a final period of rapid brain growth (i.e. the first postnatal stage of rapid growth occurs during nursing). It is during this period that different cognitive abilities are most likely to first emerge, but detecting their emergence requires a higher frequency of testing than previous studies (Bray, Gruen, et al., 2020, 2021; Lazarowski et al., 2020; Salomons et al., 2021). In this study, we define 'emergence' as the age at which puppies, as a group, perform above chance on a given task (or, for non-object choice tasks, perform the behaviour being tested on over 50% of trials). We also consider the point at which their performance on a task stops increasing as they mature.

To test the MIH more fully and document cognitive development, a high-resolution longitudinal approach is needed that repeatedly samples puppies during their final period of rapid brain growth at 8–16 weeks of age. Here, we used this approach to provide a critical test of this core developmental prediction of the MIH. We measured a range of cognitive abilities in assistance dog puppies previously shown to be critical to their training success as adults. To test the impact of socialization during rearing on cognitive maturation rates, we compared two groups of puppies raised in different ways: those intensively socialized by one main family in a raiser's home, or those extensively socialized in a puppy kindergarten setting by a team of dozens of rotating caretakers (Howell et al., 2015; Mai et al., 2021; Seksel et al., 1999). To examine the role of repeated testing on performance, we compared the final test session of our longitudinal sample to an age-matched control group only tested once. The MIH predicts that the different cognitive abilities emerge and mature at different times during this period of rapid brain growth. While rearing and repeated testing may impact development, a differentiated maturational pattern will be discernible, with independent development of skills. In contrast, the GIH suggests that improvement across tasks is uniform and tightly linked to rearing experience or exposure to each task. After accounting for rearing and test exposure, individual cognitive skills should not show significant group level patterns of emergence and maturation related to the individual's age.

METHODS

Subjects

Participants in the dog cognitive longitudinal battery (DCLB) included 113 (male:female = 57:56) puppies, all of whom are included in this analysis. Longitudinal subjects ($N = 91$) were tested on the battery multiple times at 8–20 weeks of age. Control subjects ($N = 22$) completed the battery once, typically at 16–20 weeks of age (2 controls were tested at 12–13 weeks of age due to the pandemic closures; Table 1). All puppies were Labrador retrievers, golden retrievers or Labrador $\times$ golden retriever crosses with known pedigrees from lines of dogs being bred and socialized for work as guide dogs for people with vision loss, service dogs for people with disabilities ranging from mobility challenges to PTSD, or facility dogs for professionals working in healthcare, criminal justice and educational settings (Bray, Gruen, et al., 2021). The majority of puppies came from Canine Companions (www.canine.org, $N = 97$), while the remainder came from Eyes, Ears, Nose, and Paws (www.eenp.org, $N = 14$) and Guiding Eyes for the Blind (www.guidingeyes.org, $N = 2$).

Puppies were whelped and reared with their mothers and littermates until weaning at 6–8 weeks of age, either at a professional whelping centre or in a volunteer caretaker's home, according to each organization's protocols. After weaning, and up

Table 1

Number of subjects tested on the dog cognitive longitudinal battery (DCLB) at each time point

Testing time point (age in weeks)	N longitudinal subjects	N control subjects
8–9	64	0
10–11	85	0
12–13	86	2
14–15	85	0
16–17	86	6
18–20	74	14
21	1	0

We tested 91 individuals longitudinally (i.e. across multiple time points) and 22 individuals at a single time point as controls.

to and throughout the time of testing, all puppies were then reared in one of two conditions: in a raiser's family home (intensive socialization strategy, $N = 78$), or in the Duke Puppy Kindergarten (DPK) (extensive socialization strategy, $N = 35$). In both rearing conditions, puppies were raised according to the guidelines of the organization to which they belonged. Within these guidelines their socialization to humans and same-aged puppies differed. Intensively socialized home-raised puppies primarily interacted with their raiser families and were the only puppy in the home. Typically, their raisers introduced them to new places and people on a weekly basis and socialized them with other similarly aged puppies at monthly training classes. The extensively socialized DPK-raised puppies were cared for by dozens of individuals each day. This included one to seven caretakers at a time during the day, who rotated every 2–4 h from a team of over 100 undergraduate student volunteers. DPK puppies also spent significant time daily with the other same-aged puppies in the programme. At night, DPK puppies were cared for by a smaller cohort of student volunteers, rotating every 2–3 nights, in their homes/dorms or in DPK's husbandry room. Additionally, puppies went on daily outings around Duke's campus and participated in public visiting hours most days, resulting in interactions with hundreds of new people during their time at DPK. Our extensive and intensive rearing approaches were designed after contacting six representative assistance dog organizations and reviewing their puppy raising manuals. While both rearing approaches consisted of highly positive socialization, the intensive raising strategy was more typical of a pet dog's experience, while the extensive strategy was more of an extreme in terms of the exposure to humans and other puppies among the different strategies we surveyed. While representing very different approaches to socialization, each approach may have advantages in increasing bonding, positive social behaviours and learning outcomes (Harvey et al., 2016; McEvoy et al., 2022; Vaterlaws-Whiteside & Hartmann, 2017).

Rearing group assignments were based on the logistics of volunteer availability and timing of whelping. When possible, preference for participation in the study was given to non-littermates, with the aim of including puppies from as many unique litters as possible. The 113 puppies were conceived from 69 unique dam/sire pairings, born between November 2018 and November 2023. Approximately 65% ($N = 74$) of subjects had a full sibling, and ~31% ($N = 35$) had a half sibling in the sample. Useable data were obtained from all subjects, although not all subjects completed all trials of every test session. Incomplete testing occurred occasionally due to puppies' inability to finish testing (see abort criteria) or logistical constraints (e.g. pandemic closures, illness, age-constrained timing not coinciding with staff availability). See Table 1 for the total number of subjects in the longitudinal and control sample at each age.

Ethical Note

All puppies were raised according to the guidelines of their service organizations, and all testing conducted was in accordance with these guidelines in addition to those of the Institutional Animal Care and Use Committee of Duke University (IACUC protocol A122-21-06).

Procedure

Set-up of testing area

All experiments took place from October 2019 to November 2023 in a quiet testing room at either the Duke Canine Cognition Center, or one of Canine Companions' campuses. To further reduce any distractions, a white noise machine was used during all testing. The testing area was approximately 2.4×3 m with a 1.2×1.8 m testing mat centred inside (Fig. 1). A short fence was placed around the testing area to help limit distractions and allow puppies to focus on the experimental manipulations and rewards. Water was available ad libitum via a bowl of freshwater placed in the testing area behind the start box (see Fig. 1). All testing sessions were filmed with cameras positioned to capture the behaviours of the subject and experimenter for each test. Depending on the test, this required two or three video cameras (Sony Handycam HDR-CX405 or similar). Cameras were always mounted on tripods and positioned outside the testing area.

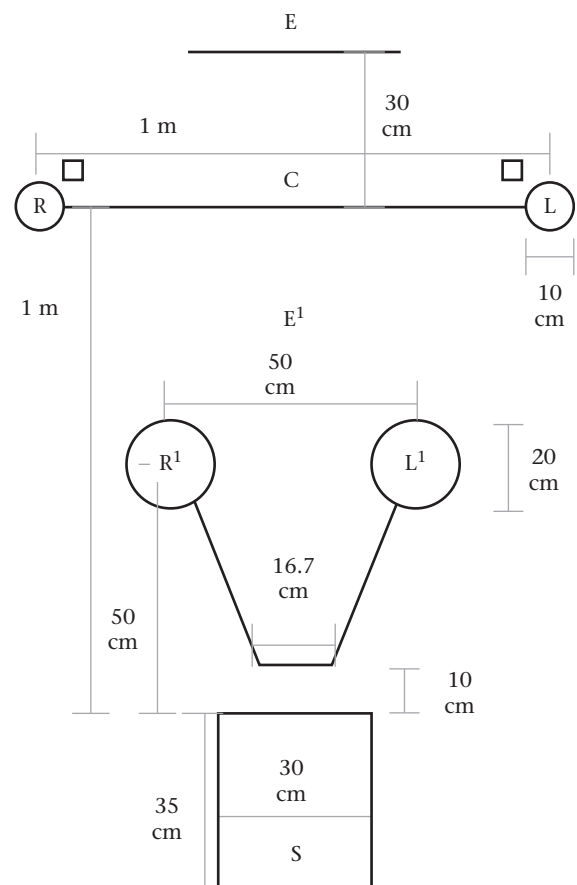


Figure 1. DCLB cognitive testing mat. Letters indicate key locations: puppy start box (S), experimenter positions (E, E¹), and possible bait locations (right (R), centre (C), left (L), R¹, L¹).

Testing protocol

Subjects were given a few minutes to explore and acclimate to the testing room and experimenters before testing began. An experimenter conducted the tests, and a handler gently positioned and held the subject in the starting position (i.e. in the starting box; see Fig. 1). Subjects began all test trials sitting in the starting box facing the experimenter. The handler sat or kneeled behind the subject, gently holding the subject in position either by the collar or by placing a hand on either side of the chest. The experimenter gained the subject's attention, carried out the experimental manipulation and indicated to the handler to release the subject by saying 'Okay'. In timed trials, the handler also started the timer once the experimenter said 'Okay'. The handler released the subject by gently removing their hand(s), making sure to remove both hands evenly to avoid biasing the subject's direction. To limit unintentional influence on the subject's choices, when not conducting the experimental manipulation, the experimenter assumed a 'resting position' where she kneeled and looked down at her lap/knees with her hands behind her back. Whenever possible, the same person played the role of the experimenter for a given puppy across sessions. However, different people played the roles of the experimenter and the handler across puppies.

In each test, subjects completed tasks to obtain food rewards. One piece of puppy kibble was typically used for each trial's reward. Subjects were given a set amount of time to search for the reward or otherwise participate in each trial (see procedures described for each test below). When a subject did not participate before the time expired ('no-choice') twice in a row, they were refamiliarized with the test procedure. If the puppy did not engage in refamiliarization trials, made a total of four no-choices, or at any time exhibited signs of significant stress (including excessive whining, barking, escape behaviour or defecation), the session was aborted and attempted again after a break of at least 30 min. Additional task-specific choice and abort criteria are described in the Supplementary Methods.

Cognitive tasks in the DCLB were chosen based on those of previous research batteries (dog cognition test battery, DCTB: MacLean, Herrmann, et al., 2017; dog cognitive development battery, DCDB: Bray et al., 2020; also see Supplementary Material), which have been associated with training success and/or significant development or individual stability from puppyhood to adulthood (Bray et al., 2019; Bray, Gruen, et al., 2021; Bray, Otto, et al., 2021; MacLean & Hare, 2018).

The DCLB consists of two warm-up sessions, nine cognitive tasks and four temperament measures as part of a larger longitudinal research programme. In the present study, we report only results of the cognitive tasks to test the predictions of our hypotheses. Specifically, cognitive tests were meant to correspond to nonsocial domains of cognition (i.e. memory: working memory distraction and delay tasks; physical reasoning: causal reasoning task; sensory perception: auditory and odour discrimination tasks; inhibition: cylinder task) as well as social domains of cognition (i.e. communicative intentions: momentary pointing, marker gesture tasks; referential communication: unsolvable task). Cognitive testing took place over two sessions (Table 2), which were completed across 1–2 days based on puppy and raiser availability, with temperament tasks typically being run between the two sessions.

Design

We used a within-subject, longitudinal design. The study was run as a battery of tests, in which subjects participated in all the cognitive tests in the same order. Longitudinal subjects were tested on the DCLB every 2 weeks for 10 weeks, from ages 8–18,

Table 2
Order of dog cognitive longitudinal battery (DCLB) cognitive testing tasks

Session 1	Session 2
Unsolvable (eye contact)	Cylinder (inhibitory control and reversal learning)
Bowl choice warm-ups	Bowl choice warm-ups
Marker gesture	Momentary pointing gesture
Working memory (distraction)	Working memory (delay)
Causal reasoning	Auditory discrimination
	Odour discrimination

9–19 or 10–20 weeks (typically completing six test sessions, but some participants missed sessions due to availability constraints). To assess the impact of repeated testing on the longitudinal sample, control subjects were tested on the DCLB only once. In tasks where food was hidden in one of two locations, the order across trials of the 'correct' baited location was the same for all subjects (following Herrmann et al., 2007; MacLean & Hare, 2018). Reward placement was pseudorandomized across the different tasks in the battery using the constraint that the reward was hidden on the right and left an equal number of times in each task and never on the same side more than twice in a row. The dependent variable for each test was generally whether the puppy correctly solved the test (with the exception of the unsolvable test, which measured seconds of eye contact) and are described in detail in the Supplementary Methods. The independent variables of interest included age (in weeks), rearing strategy (DPK versus home-raised) and prior testing experience (number of prior testing sessions).

DCLB Task Protocols

In the bowl choice warm-ups, the experimenter visibly placed food in one of two bowls (in positions right and left; Fig. 1) so that the puppy became familiar with choosing, approaching and eating from the bowls.

In the marker gesture task, the experimenter hid food in one of the two bowls behind an occluder. Once the occluder was removed and the bowls were revealed, the experimenter used ostensive cues (i.e. attempted to make eye contact and said, 'Puppy look!' in a high-pitched voice) while placing a small yellow wooden block next to the correct bowl to indicate the hiding location.

In the working memory tasks, the puppy was allowed to see the bowl into which the food was placed, then released to search for it after 20 s. In the 'delay' condition, the experimenter remained motionless and silent during the delay; in the 'distraction' condition, the experimenter engaged the puppy in play with a toy during the delay.

In the auditory and odour discrimination tasks, the puppy was asked to determine which of two close hiding locations, to the right and left of the experimenter (positions R¹ and L¹, Fig. 1), contained the food based on these senses. In the auditory task, a piece of dry kibble was audibly dropped into one of two metal bowls, while the other bowl was silently sham-baited. In the odour task, two plastic tubes with openings covered by mesh fabric were presented to the puppy to sniff, only one of which contained food.

In the cylinder task, puppies retrieved food from a transparent cylinder; to succeed in the inhibitory control condition, they had to approach one of the open ends of the cylinder rather than touch the cylinder walls, through which they could see the food. In the reversal condition, the puppy's preferred side from previous trials was blocked; thus, to succeed in this task, they had to learn to approach the cylinder from the other side without bumping the cylinder.

In the unsolvable (eye contact) task, food was placed in a sealed transparent container, and the experimenter timed how much eye contact the puppy made with them while the food was inaccessible.

In the causal reasoning task, the experimenter placed a cloth over a bowl containing food in one hiding location, while an identical cloth was laid flat on the floor in the other hiding location, and the puppy had to choose a location.

The momentary pointing task was identical to the marker gesture task, except that instead of placing a marker, the experimenter indicated the hiding location by pointing with the contralateral arm, held the point for 3 s, then returned to resting position before the puppy was released to make a choice.

Detailed protocols for each task are provided in the Supplementary Material.

Scoring and Analysis

Choice and scoring definitions for each test are included in the Supplementary Methods. All trials were recorded from an angle (or multiple angles) that captured the subject's response as well as the experimenter. Subject responses were live-coded by the experimenter after each trial for all cognitive measures. In a minority of cases the experimenter was unable to live-code (e.g. due to stopwatch malfunction) and later coded the trial from video. Approximately 20% of all sessions (chosen pseudorandomly ensuring that subjects from both rearing strategies as well as controls were included) were independently coded from video to determine reliability with the live coding. Reliability was excellent on all tests: marker gesture (Cohen's $\kappa = 0.98$, $N = 405$, $P < 0.001$); momentary pointing (Cohen's $\kappa = 0.95$, $N = 396$, $P < 0.001$); working memory (delay) (Cohen's $\kappa = 0.99$, $N = 410$, $P < 0.001$); working memory (distraction) (Cohen's $\kappa = 0.98$, $N = 400$, $P < 0.001$); causal reasoning (Cohen's $\kappa = 0.97$, $N = 397$, $P < 0.001$); auditory discrimination (Cohen's $\kappa = 0.97$, $N = 397$, $P < 0.001$); odour discrimination (Cohen's $\kappa = 0.96$, $N = 390$, $P < 0.001$); cylinder (inhibitory control) (Cohen's $\kappa = 0.96$, $N = 388$, $P < 0.001$); cylinder (reversal learning) (Cohen's $\kappa = 0.94$, $N = 386$, $P < 0.001$); unsolvable (eye contact) (Pearson correlation: $r_{392} = 0.96$, $P < 0.001$).

Analysis was completed in RStudio version 2023.06.1+524. Mixed effects logistic and linear regression models (depending on the test: logistic for binary dependent variables and linear for continuous) were run, including age (in weeks), sex, rounds of previous test battery experience and rearing strategy (DPK versus home-raised) as fixed effect variables. Individual puppy ID was included as a random effect in all models to control for repeated testing. These models were compared using Akaike information criterion (AIC) scores to determine which variables significantly impacted the outcome of each test and the nature of their effects. Age and experience can be differentiated in these analyses, in part, because the start of biweekly testing was not perfectly synchronized across subjects. Some puppies started testing when they were 8 weeks ($N = 32$), 9 weeks ($N = 30$) or 10–14 weeks ($N = 29$) of age and were tested biweekly from their first test date (see Supplementary Table S15). In addition, some puppies did not complete all testing within each session (e.g. following the abort criteria) or, due to constraints on availability, were tested on fewer than six longitudinal time points. This meant that puppies of the same age did not always have the same amount of experience, providing us an opportunity to examine the influence of age and experience separately within the longitudinal sample.

To examine the effect of repeated testing on performance further, we conducted two-tailed Mann–Whitney U tests to compare the performance of longitudinal subjects in their final (fifth or sixth) testing session with same-age (17–21 weeks)

control subjects in their sole testing session (longitudinal mean age = 18.52 weeks; control mean age = 18.55 weeks; two-tailed Welch's t test showed no significant difference in age: $t_{25.07} = 0.10$, $P = 0.92$). To determine age of emergence of each skill, we ran Z tests for each age group against the null hypothesis of 50% correct (or the 'proportion of trials looked' for the unsolvable task) (intercept = 0) using a mixed effect logistic regression model.

For skills on which puppies reached above chance performance at any point during the 8–21-week age window and that also showed a positive linear relationship with age, we performed segmented regression analysis and determined a breakpoint using the Segmented package (Muggeo, 2008) with age as the independent variable and the proportion of trials correct as the dependent variable. We defined the breakpoint as the age at which there was a significant change in the slope of the performance in relation to age.

RESULTS

Marker Gesture Task

Age was a significant predictor variable in all models in which it was considered, while rearing strategy and sex were never significant. Experience was a significant predictor variable when entered into the model alone but became insignificant when added to models that also included age. The model with the lowest AIC score (876.116) contained only age as a predictor variable ($\beta_{\text{age}} = 0.114$, $SE = 0.023$, $P < 0.001$) (Supplementary Table S2).

A two-tailed Mann–Whitney U test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects in their last testing session (only those aged 17–21 weeks old, $N = 74$, mean $\pm$ SD = 0.91 ± 0.28) to that of control subjects in the same age range ($N = 20$, mean $\pm$ SD = 0.91 ± 0.28) showed that they performed similarly ($W = 11844$, $P = 0.99$; Table 3, Supplementary Fig. S1).

Figure 2a (by weeks) and Supplementary Fig. S2 (by days) show all longitudinal puppies' ($N = 91$) performance on the marker gesture task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S3 shows performance of the same longitudinal puppies on the marker gesture task grouped by rearing strategy.

Working Memory (Delay) Task

Age and experience were significant predictor variables in all the models in which they were entered. Rearing was a significant predictor variable in the model in which it was entered alone and in the age + rearing model. Sex was only a significant predictor in the model in which it was entered alone. The model with the lowest AIC score (1072.844) contained age and experience as predictor variables ($\beta_{\text{age}} = 0.201$, $SE = 0.051$, $P < 0.001$; $\beta_{\text{experience}} = -0.300$, $SE = 0.103$, $P < 0.001$; Supplementary Table S3).

A two-tailed Mann–Whitney U test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 76$, mean $\pm$ SD = 0.86 ± 0.35) to that of control subjects in the same age range ($N = 19$, mean $\pm$ SD = 0.95 ± 0.22) showed that the controls performed significantly better ($W = 12578$, $P < 0.05$; Table 3, Supplementary Fig. S4).

Figure 2b and Supplementary Fig. S5 show all longitudinal puppies' ($N = 91$) performance on the working memory with delay task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary

Table 3
Differences between longitudinal puppies in their final testing session and same age (17–21 weeks) control puppies

Variable	Units	Longitudinal mean ± SD	Control mean ± SD	t	df	P
Marker gesture	Proportion trials correct	0.91±0.28	0.91±0.28	0.009	124.75	0.99
Working memory (delay)	Proportion trials correct	0.86±0.35	0.95±0.22	2.7208	176.78	<0.01
Working memory (distraction)	Proportion trials correct	0.79±0.41	0.79±0.41	-0.06	124.08	0.95
Auditory discrimination	Proportion trials correct	0.57±0.50	0.69±0.47	2.03	130.77	<0.05
Odour discrimination	Proportion trials correct	0.50±0.50	0.56±0.50	0.90	123.05	0.37
Cylinder (inhibitory control)	Proportion trials correct	0.94±0.23	0.68±0.47	-4.87	89.99	<0.001
Cylinder (reversal learning)	Proportion trials correct	0.62±0.49	0.28±0.45	-5.98	133.18	<0.001
Unsolvable (looking time)	No. of seconds	5.34±6.50	3.07±4.46	-3.59	184.91	<0.001
Unsolvable (prop. looked)	Proportion trials looked	0.81±0.39	0.60±0.49	-3.47	109.44	<0.001
Causal reasoning	Proportion trials correct	0.89±0.31	0.43±0.50	-6.25	59.68	<0.001
Momentary pointing	Proportion trials correct	0.51±0.50	0.48±0.50	-0.45	117.66	0.65

Bolded means for each measure indicate when a group performed significantly higher. Note: eye contact was measured and is reported here in two ways (i.e. total looking time and proportion of trials where looking occurred) for the unsolvable task.

Fig. S6 shows performance of the puppies grouped by rearing strategy.

Working Memory (Distraction) Task

Age was a significant predictor variable in all models in which it was entered. Experience was a significant predictor variable when entered into the model alone and with age, but not in the model with age and rearing. Rearing was not a significant predictor variable in any of the models in which it was entered. Sex was a significant predictor only when entered into the model with all other variables. The model with the lowest AIC score (1222.845) included age and rearing as predictor variables ($\beta_{\text{age}} = 0.091$, $\text{SE} = 0.016$, $P < 0.001$; $\beta_{\text{rearing}} = -0.311$, $\text{SE} = 0.204$, NS; [Supplementary Table S4](#)).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 74$, mean ± SD = 0.79 ± 0.41) to that of control subjects in the same age range ($N = 20$, mean ± SD = 0.79 ± 0.41) showed that both groups performed similarly ($W = 11804$, $P = 0.95$; [Table 3](#), [Supplementary Fig. S7](#)).

[Figure 2c](#) and [Supplementary Fig. S8](#) show all longitudinal puppies' ($N = 91$) performance on the working memory with distraction task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' ([Wickham, 2016](#)), and [Supplementary Fig. S9](#) shows performance of the same puppies grouped by rearing strategy.

Auditory Discrimination Task

Age was a significant predictor variable in the age alone model and the age + rearing model but became insignificant when experience was entered into the models. Experience was a significant predictor in all the models in which it was entered. Rearing and sex were not significant predictor variables in any of the models in which it was entered. The model with the lowest AIC score (1274.966) included only experience as a predictor variable ($\beta_{\text{experience}} = -0.135$, $\text{SE} = 0.029$, $P < 0.001$; [Supplementary Table S5](#)).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 75$, mean ± SD = 0.57 ± 0.50) to that of control subjects in the same age range ($N = 20$, mean ± SD = 0.69 ± 0.47) showed that both groups performed similarly ($W = 13450$, $P = 0.05$; [Table 3](#), [Supplementary Fig. S10](#)).

[Figure 2d](#) and [Supplementary Fig. S11](#) show all longitudinal puppies' ($N = 91$) performance on the auditory discrimination task across the study period by weeks and days using the 'geom_smooth' function, method = 'loess' ([Wickham, 2016](#)), and [Supplementary Fig. S12](#) shows performance of the same puppies grouped by rearing strategy.

Odour Discrimination Task

Age was a significant predictor variable in the age alone model and the age + rearing model. Experience was a significant predictor variable when entered into the model alone but became insignificant when added to models that also included age. Rearing and sex were not significant predictor variables in any of the models in which they were entered. The model with the lowest AIC score (1225.646) included age and rearing as predictor variables ($\beta_{\text{age}} = -0.029$, $\text{SE} = 0.013$, $P < 0.01$; $\beta_{\text{rearing}} = 0.146$, $\text{SE} = 0.099$, NS; [Supplementary Table S6](#)).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 74$, mean ± SD = 0.50 ± 0.50) to that of control subjects in the same age range ($N = 19$, mean ± SD = 0.56 ± 0.50) showed that both groups performed similarly ($W = 12358$, $P = 0.37$; [Table 3](#), [Supplementary Fig. S13](#)).

[Figure 2e](#) and [Supplementary Fig. S14](#) show all longitudinal ($N = 91$) puppies' performance on the odour discrimination task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' ([Wickham, 2016](#)), and [Supplementary Fig. S15](#) shows performance of the same puppies grouped by rearing strategy.

Cylinder Task

Inhibitory control trials

Age and experience were both significant predictor variables in all the models in which they were entered. Rearing was not a significant predictor variable in any of the models. Sex was only significant in the model in which it was entered alone. The model with the lowest AIC score (957.181) included age and experience ($\beta_{\text{age}} = 0.084$, $\text{SE} = 0.028$, $P < 0.001$; $\beta_{\text{experience}} = 0.924$, $\text{SE} = 0.077$, $P < 0.001$; [Supplementary Table S7](#)).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which a clear route was taken to an open side without touching the transparent cylinder wall by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 75$, mean ± SD = 0.94 ± 0.23) to that of control subjects in the same

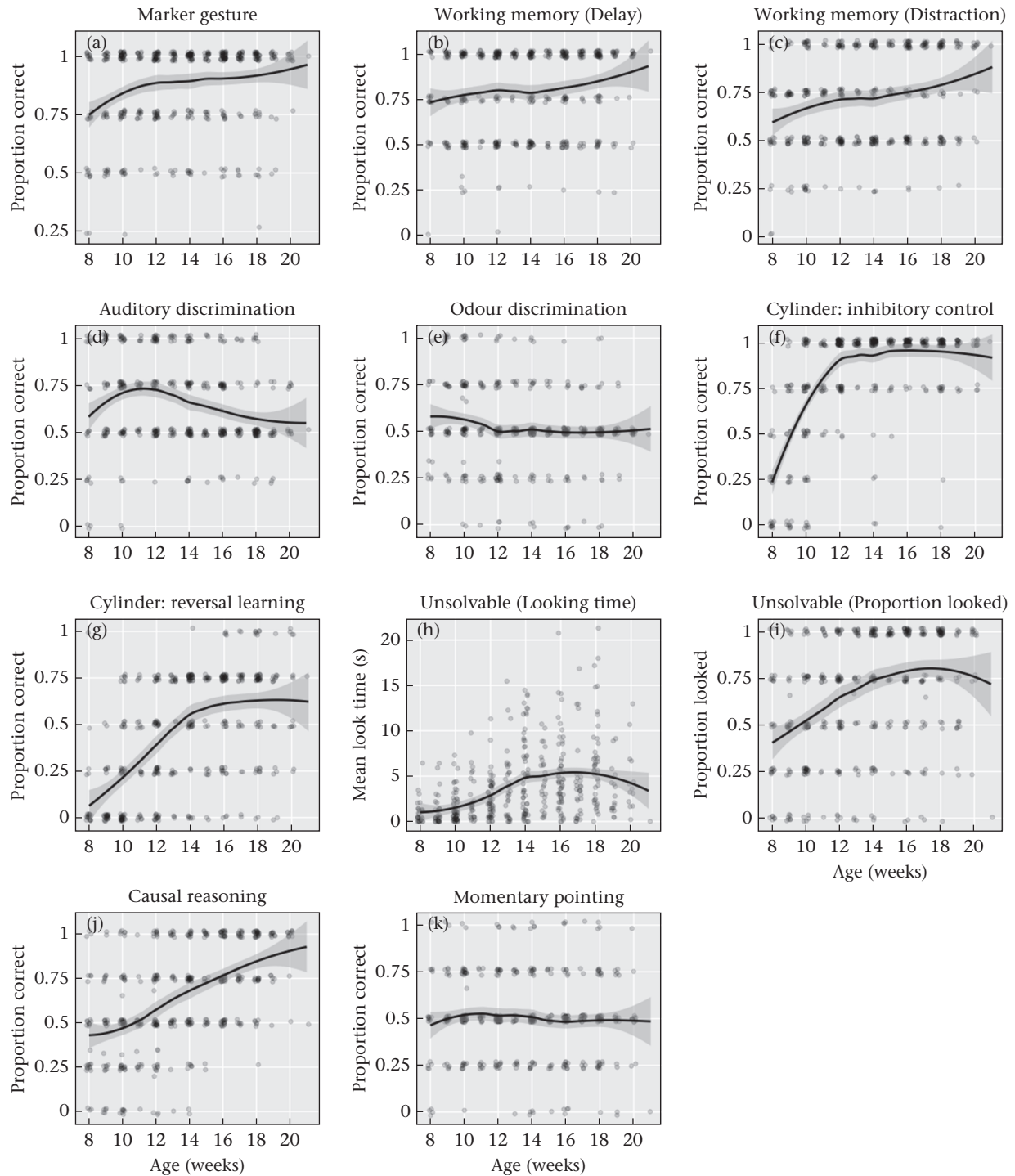


Figure 2. Performance on each task by age in weeks.

age range ($N = 20$, mean $\pm$ SD = 0.68 ± 0.47) showed that the longitudinal puppies performed significantly better than the control puppies ($W = 8820$, $P < 0.0001$; Table 3, Supplementary Fig. S16).

Figure 2f and Supplementary Fig. S17 show all longitudinal puppies' ($N = 91$) performance on the inhibitory control trials across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S18 shows performance of the same puppies grouped by rearing strategy.

Reversal learning trials

Age and experience were both significant predictor variables in all the models in which they were entered. Rearing was not a significant predictor variable in any of the models. Sex was only significant in the model in which it was entered alone. The model with the lowest AIC score (1364.161) included age, experience and sex ($\beta_{\text{age}} = 0.061$, SE = 0.026, $P < 0.01$; $\beta_{\text{experience}} = 0.410$, SE = 0.052, $P < 0.001$; $\beta_{\text{sex}} = 0.194$, SE = 0.129, NS; Supplementary Table S8).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which a clear route was taken to an open side without touching the transparent cylinder wall by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 75$, mean $\pm$ SD = 0.62 ± 0.49) to that of control subjects in the same age range ($N = 20$, mean $\pm$ SD = 0.28 ± 0.45) showed that the longitudinal puppies performed significantly better than the control puppies ($W = 7787$, $P < 0.0001$; Table 3, Supplementary Fig. S19).

Figure 2g and Supplementary Fig. S20 show all puppies' performance on the inhibitory control reversal task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S21 shows performance of the same puppies grouped by rearing strategy.

Unsolvable (Eye Contact) Task

Total looking time

Age, experience and sex were significant predictor variables in all the models in which they were entered. Rearing was not a significant predictor variable in any of the models. The model with the lowest AIC score (2630.829) included age, experience and sex ($\beta_{\text{age}} = 0.209$, SE = 0.074, $P < 0.001$; $\beta_{\text{experience}} = 0.512$, SE = 0.150, $P < 0.001$; $\beta_{\text{sex}} = 1.167$, SE = 0.402, $P < 0.001$; Supplementary Table S9).

A two-tailed Mann–Whitney *U* test comparing the looking times (s) of the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 74$, mean $\pm$ SD = 5.34 ± 6.50) to that of control subjects in the same age range ($N = 20$, mean $\pm$ SD = 3.07 ± 4.46) showed that the longitudinal puppies made significantly longer eye contact than the controls ($W = 8343.5$, $P < 0.001$; Table 3, Supplementary Fig. S22).

Figure 2h and Supplementary Fig. S23 show all longitudinal puppies' ($N = 91$) mean looking times across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S24 shows mean looking times for the same puppies grouped by rearing strategy.

Proportion of trials looked

Age, experience and sex were significant predictor variables in all the models in which they were entered. Rearing was not a significant predictor variable in any of the models. The model with the lowest AIC score (1428.953) included age and experience as predictor variables ($\beta_{\text{age}} = 0.087$, SE = 0.031, $P < 0.001$; $\beta_{\text{experience}} = 0.220$, SE = 0.063, $P < 0.001$; $\beta_{\text{sex}} = 0.386$, SE = 0.187, $P < 0.01$; Supplementary Table S10).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the longitudinal subjects made any eye contact with the experimenter during their last testing session (only those aged 17–21 weeks old, $N = 74$, mean $\pm$ SD = 0.81 ± 0.39) to that of control subjects in the same age range ($N = 20$, mean $\pm$ SD = 0.60 ± 0.49) showed that the longitudinal puppies made eye contact in a significantly higher proportion of trials than control puppies ($W = 8904$, $P < 0.001$; Table 3, Supplementary Fig. S25).

Figure 2i and Supplementary Fig. S26 show all longitudinal puppies' ($N = 91$) proportion of trials in which they made eye contact across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S27 shows the proportion of trials in which the same puppies made eye contact grouped by rearing strategy.

Causal Reasoning Task

Age was a significant predictor variable in the age alone model and the age + rearing model. Experience was a significant

predictor in all the models in which it was entered. Rearing was a significant predictor variable in all the models in which it was entered. Sex was not a significant predictor variable in any of the models in which it was entered. The model with the lowest AIC score (1241.111) included age, experience, rearing and sex as predictor variables ($\beta_{\text{age}} = 0.003$, SE = 0.024, NS; $\beta_{\text{rearing}} = -0.310$, SE = 0.134, $P < 0.01$; $\beta_{\text{experience}} = 0.489$, SE = 0.052, $P < 0.001$; $\beta_{\text{sex}} = -0.177$, SE = 0.125, NS; Supplementary Table S11).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 74$, mean $\pm$ SD = 0.88 ± 0.32) to that of control subjects in the same age range ($N = 19$, mean $\pm$ SD = 0.41 ± 0.50) showed that the longitudinal puppies performed significantly better than the control puppies ($W = 5900.5$, $P < 0.0001$; Table 3, Supplementary Fig. S28).

Figure 2j and Supplementary Fig. S29 show all longitudinal puppies' ($N = 91$) performance on the causal reasoning task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S30 shows performance of the same puppies grouped by rearing strategy (intensive $N = 35$; extensive $N = 77$).

Momentary Pointing Task

None of the covariates (age, experience, rearing or sex) was a significant predictor in any of the models. The model with the lowest AIC score (1269.087) included only rearing strategy as a predictor variable ($\beta_{\text{rearing}} = 0.071$, SE = 0.098, NS; Supplementary Table S12).

A two-tailed Mann–Whitney *U* test comparing the proportion of trials in which the baited bowl was chosen by the longitudinal subjects on their last testing session (only those aged 17–21 weeks old, $N = 75$, mean $\pm$ SD = 0.51 ± 0.50) to that of control subjects in the same age range ($N = 20$, mean $\pm$ SD = 0.48 ± 0.50) did not detect a significant difference in performance ($W = 11210$, $P = 0.65$; Table 3, Supplementary Fig. S31).

Figure 2k and Supplementary Fig. S32 show all longitudinal puppies' ($N = 91$) performance on the momentary pointing gesture task across the 8–21-week age window using the 'geom_smooth' function, method = 'loess' (Wickham, 2016), and Supplementary Fig. S33 shows performance of the same puppies grouped by rearing strategy (extensive $N = 35$; intensive $N = 77$). See Table 4 for summary of the age of emergence for each cognitive tasks based on criteria of either 50% of sample at each age performing above chance or making correct response on at least half of trials.

A summary of the best best-fit models is presented in Supplementary Table S13. A summary of the effects of age, rearing, and experience for each task is presented in Table 5.

Order of Rapid Development Completion (Breakpoint Analysis)

Of the skills that were fitted with segmented regression lines and breakpoints (see Methods), four (marker gesture, cylinder inhibitory control, cylinder reversal learning and unsolvable (eye contact)) were found to have a pattern of rapidly increasing performance, with a breakpoint where performance levelled out during the 8–21-week age window. One skill (causal reasoning) did not show this pattern, as performance instead continued to increase steadily across the age window. As such, the only breakpoint determined was very early, after a short period of decrease before the rapid increase began, and is likely an artefact of the relatively small sample size at the very young end of the age window (~8 weeks). Both of the working memory skills (delay and

Table 4
Order of emergence of cognitive skills

	8–9 weeks	10–11 weeks	12–13 weeks	14–15 weeks	16–17 weeks	18–20 weeks
Marker gesture	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a
Working memory (delay)	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a
Working memory (distraction)	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a
Auditory discrimination	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a
Odour discrimination	** ^a	** ^a	NS	NS	NS	NS
Cylinder (inhibitory control)	** ^b	*** ^a	*** ^a	*** ^a	*** ^a	*** ^a
Unsolvable (proportion looked)	NS	* ^a	*** ^a	*** ^a	*** ^a	*** ^a
Causal reasoning	NS	NS	*** ^a	*** ^a	*** ^a	*** ^a
Cylinder (reversal learning)	*** ^b	*** ^b	* ^b	* ^a	* ^a	* ^a
Momentary pointing	NS	NS	NS	NS	NS	NS

Performance on each task during biweekly testing. The 50% emergence criterion represents chance for all tasks except for the cylinder and unsolvable tasks, for which the emergence criterion represents when puppies, as a group, showed the skill on more than half the trials. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. 'NS' indicates no significant difference from 50% correct.

^a Puppies, as a group, met emergence criteria (>50% of trials correct) for the skill at the significance level indicated by the asterisks in bold.

^b Performance was significantly below 50% correct at the significance level indicated by the asterisks.

Table 5
Summary of age, rearing and experience effects on dog cognitive longitudinal battery (DCLB) cognitive tasks

Task	Age effect?	Rearing effect?	Experience effect?
Marker gesture	+	NS	NS
Working memory (delay)	+	NS	—
Working memory (distraction)	+	NS	NS
Auditory discrimination	+	NS	—
Odour discrimination	+	NS	NS
Cylinder inhibitory control	+	NS	+
Cylinder reversal learning	+	NS	+
Eye contact	+	NS	+
Causal reasoning	+	DPK>home	+
Momentary pointing	NS	NS	NS

'+' indicates a significant positive effect on performance; '—' indicates a significant negative effect on performance; 'NS' indicates no significant effect on performance. DPK: Duke Puppy Kindergarten.

distraction) showed a slow, steady increase across most of the age window, with a breakpoint after which performance seemed to increase more rapidly towards the older end of the window. However, this could be due to a relatively small sample size at the older end (~20 weeks) as well.

The four skills for which development reached a plateau during the tested age window are shown together (Fig. 3), to compare the ages at which the performance on these skills began to level out, indicating the end of rapid development. The piecewise linear regressions and breakpoints for each skill are given in the Supplementary Material (Figs S34–S41).

DISCUSSION

Our findings provide strong support for the multiple intelligences hypothesis. Nine out of the 10 skills measured were observed to emerge as puppies were completing their final stage of rapid brain growth before 16 weeks of age. The pattern of emergence was not uniform across cognitive tasks, nor was it tightly linked to previous experiences, as predicted by unidimensional learning or a general intelligence mechanism. Rather, cognitive skills emerged at different time points and developed at different rates while being differentially impacted by experience (Tables 4–5). Development took a variety of paths, with skills developing in patterns ranging from rapid (inhibitory control) to gradual (causal reasoning) (Fig. 2). Developmental patterns differed between tasks (Fig. 2), even though each task had four trials each and similar rewards, and individuals were tested on the

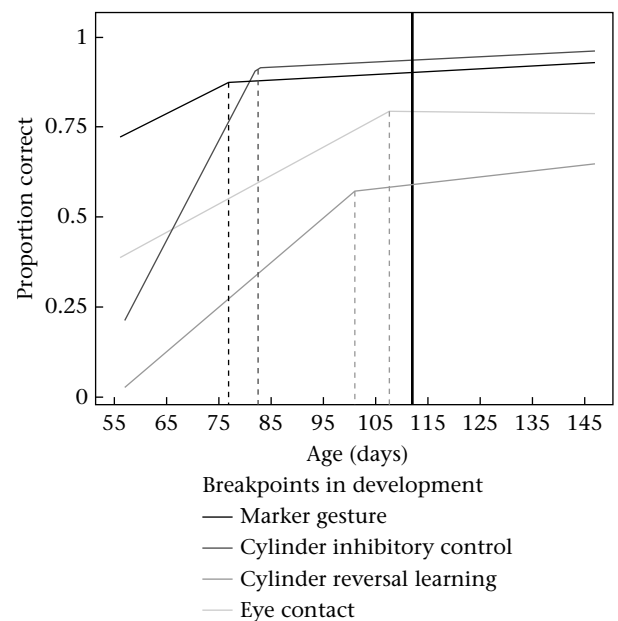


Figure 3. Combined plot of segmented regression lines for the four tasks that exhibited the pattern of rapid development completion within the 8–21-week age window. The solid line at 112 days represents 16 weeks or the end of the most rapid brain growth in puppies.

same schedule. Differential rearing had almost no impact on cognitive ontogeny. Extensive exposure to same-aged puppies and novel people and places on a college campus did not alter the development trajectory of social cognition in comparison to puppies intensively reared as part of a human family. Only a single nonsocial measure (causal reasoning) was impacted by rearing, with no ready explanation for this relationship. Repeated testing of the longitudinal puppies did not consistently affect their performance across tasks when compared to same-aged controls who were only tested once. While longitudinal dogs outperformed controls on some measures, they performed similarly or less skilfully on others. The overall pattern is most consistent with the MIH. Different cognitive abilities emerged and developed at different ages and rates but showed a highly similar maturation pattern even when puppies were intentionally raised two different ways. Experience impacted development, but across different cognitive abilities, maturational patterns varied in their plasticity, with some skills being impacted more by repeated testing than others (Table 5).

Our findings provide the highest resolution description of dog cognitive development currently available. The subjects all came from closed breeding populations that have been bred for assistance work for decades. All subjects were raised similarly prior to 8 weeks of age and were tested using the same procedures (with high reliability observed across all measures). Despite these similarities, the DCLB revealed developmental variability in the emergence of cognitive skills. By defining 'emergence' as the longitudinal subjects' group level performance exceeding 50% of trials correct, we observed the following patterns (Table 4, Supplementary Table S16). First, when using a novel gesture (marker), working memory and sensory discrimination were present at 8–9 weeks, inhibitory control and eye contact (proportion of trials looked) emerged at 10–11 weeks, causal reasoning emerged at 12–13 weeks and reversal learning emerged at 14–15 weeks. Only momentary pointing failed to emerge during the developmental window we examined (Table 4).

Second, by testing whether puppies reached a performance maximum followed by a plateau demonstrates how rates of emergence differed. For the marker gesture, inhibitory control, reversal learning and eye contact, the breakpoint analysis found the predicted change from a positive to near-zero slope, with each occurring at a different time before 16 weeks of age (Fig. 3). Supporting the idea that these represent maturational changes, a subject's age was also a significant explanatory variable in the model comparisons, with a positive beta coefficient in the best-fit model for all four of these tasks. In contrast, working memory and causal reasoning emerged in a different pattern. Both skills showed steady development across test sessions, with the best-fit models all including age as a predictor variable. However, puppies did not reach a ceiling or plateau within the developmental window tested. The observed developmental trajectory only led to gradual improvement that likely continued after our final test period at 18–20 weeks. Meanwhile, both the sensory tasks showed regression instead of improvement over test sessions. The best-fit model for odour only included age, while that for auditory discrimination only included experience, with both having a significant negative beta coefficient on these respective variables. This drop in performance may have occurred as dogs became more reliant on visual information as their altricial visual cortex matured, and/or become expectant of visual cues as they gained experience with the other tests. The final pattern observed was a lack of development, with puppies showing no ability to use a momentary pointing gesture throughout the developmental window examined.

The DCLB also helped reveal what types of experience impact cognitive development and how that experience differentially impacts performance on the various tasks. First, the two rearing strategies had limited impact on cognitive emergence or the speed of development. The approaches were largely designed based on strategies that different assistance dog groups employ and differed on social variables previously shown to impact adult social behaviour (e.g. Harvey et al., 2016; Vaterlaws-Whiteside & Hartmann, 2017). Perhaps most surprising was the lack of difference in performance on social tasks, despite half the puppies being extensively exposed to new people and other similarly aged puppies. The GIH predicts that, through a general learning mechanism, greater exposure to social interactions should alter the maturational course of social skills, but instead, we found that only the nonsocial causal reasoning task showed a significant effect of rearing strategy on performance. Second, comparing control dogs to the longitudinal dogs at the final testing time point (17–21 weeks) provides a powerful test of the impact of repeated testing on performance (Table 3). In 6 of 10 tasks the control dogs performed as well or better than the longitudinal sample. In cases

where longitudinal puppies were outperformed, it may have been that repeatedly completing a task they had succeeded with so early in life led to a loss of motivation and performance. For the remaining four tasks, the longitudinal dogs outperformed the same-age control dogs, making more eye contact in a higher proportion of trials and showing more inhibition, greater ability to reverse their learning and more skill in the causal reasoning task. Repeated testing from an early age likely gave the longitudinal dogs an advantage, but the control dogs did perform above chance or have adult-like performance in the eye contact and inhibition tasks (see Supplementary Table S14). This pattern suggests that when repeated testing positively impacted performance, it did not cause a skill to emerge but instead interacted with brain maturation to enhance performance. It is also difficult to use reward history to explain the cases where the longitudinal dogs outperformed the controls because the unsolvable and cylinder tasks used to measure eye contact, inhibition and reversal learning were all nondifferentially rewarded. Only the causal reasoning task was differentially rewarded, but control puppies performed as well as or better than longitudinal puppies on all the other object choice tasks in which only correct choices were rewarded. It is easiest to interpret this pattern as a signal of maturation interacting with experience.

As a first attempt to closely examine the development of dog cognition during this final period of rapid brain development, our experiment has limitations. While we observed significant development in the puppies' first test session at 8–9 weeks of age, the longitudinal puppies performed above chance on half of our tasks (memory, marker and sensory tasks; replicating previous findings from Bray, Gruen, et al., 2020, 2021; Salomons et al., 2021). This suggests that cognitive development occurring during the final period of rapid brain development tested here is a continuation of development occurring postpartum during nursing. Future research will be needed to examine the initial development of these skills before weaning (e.g. Riedel et al., 2008). Likewise, even in the final test session (i.e. typically after 20 differentially rewarded trials across five previous biweekly sessions), longitudinal puppies failed to comprehend momentary pointing. Adult dogs can reliably read this gesture (Salomons et al., 2024), and puppies from the same assistance dog population can also reliably read the same gesture when the point is held statically as they make their choice (Bray et al., 2020). The current study indicates that puppies do have the necessary working memory by this age, which suggests that perhaps they are not attending enough to the gesture to subsequently remember it, or that the cognitive load of interpreting the cue combined with remembering it is too high. This raises the question of when the tendency to remember more subtle communicative gestures first emerges. The DCLB is also largely designed based on tasks associated with or even predictive of adult training outcomes in working dogs (e.g. Bray et al., 2019; Bray, Otto, et al., 2021; Bray, Sammel, Cheney, et al., 2017; MacLean & Hare, 2018). Future research can be designed to examine the development of a larger range of social and nonsocial cognitive abilities. Particularly interesting might be the set of tasks previously shown to have a more human-like pattern in adult dogs than in great apes (MacLean, Herrmann, et al., 2017). As assistance dogs, our subjects were also only sampled from one subpopulation of dogs. These dogs are known to differ from pet dogs in having xenophilic responses towards strange people and higher circulating baseline concentrations of oxytocin (Bray et al., 2019; MacLean, Gesquiere, et al., 2017). Many of the developmental patterns observed here may be specific to this population. A future priority will be to use the DCLB with a range of other dog populations to test for generalizability.

While recognizing the need for further research, our current findings have important implications for a puppy's potential to learn during development. Understanding when different cognitive abilities emerge informs when different types of skills are most likely to be successfully learned. Puppies will likely succeed when tasks requiring self-control are introduced after 12 weeks of age, or even 14 weeks of age for those tasks requiring the most self-control (such as reversal learning). Recognizing the different speeds with which different abilities develop also hints at how quickly puppies might learn different skills. Tasks that require self-control may rapidly increase after 12 weeks of age, while problems recruiting working memory may only gradually improve. Evaluating the impact of experience also suggests ways that social interactions may or may not enhance social learning in puppies. While our different rearing strategies produced no cognitive differences, participating in the unsolvable task for a few minutes every other week almost doubled the amount of eye contact longitudinal puppies made in this task. This type of task might be used to increase eye contact and social bonding between humans and dogs. Future research can examine whether early exposure to the unsolvable tasks leads to more eye contact in other contexts as dogs reach adulthood. It is also likely that our two rearing strategies both allowed for typical cognitive development with each strategy providing adequate levels of socialization. However, while it would be unethical to test experimentally, it is quite possible that there is a minimum threshold for socialization, and that inadequate socialization would disrupt the developmental pattern seen here. There is also the potential to retest our subjects on the DCLB when they reach adulthood, to further examine the stability of cognitive performance across the life span. Critically, this will include testing whether individual differences in cognitive performance during puppies' development are related to their professional training outcomes as adults. It may be that evaluations of each cognitive skill conducted soon after that skill has emerged may lead to the strongest predictions, which can inform the timing of cognitive evaluations of assistance dog puppies in the future. Previous work also suggests that combining cognitive and temperament measures provides increased predictive power of adult training success (Bray, Sammel, Cheney, et al., 2017; Bray, Sammel, Seyfarth, et al., 2017; Lazarowski et al., 2019; Smith et al., 2024). Combining developmental measures of temperament with our cognitive measures may again provide the most powerful way to anticipate training outcomes before an adult dog even enters professional training.

Taken together, and despite the limitations, the results further support prior research demonstrating different cognitive domains in dogs. The cognitive abilities evaluated by the DCLB develop separately, at different rates, and during a final period of rapid brain growth in puppies. Cognitive skills do not uniformly improve in synchrony with age, or in response to different types of experience as predicted by a unidimensional, generalized mechanism. Age was repeatedly the defining factor explaining patterns of performance, but the comparison of the longitudinal and control dogs showed that experience played an important role in some cases as well. Based on our findings, the question is not whether cognition and the developing brain is impacted by experience, but rather, to what degree and by which types of experience? Some types of cognition appear to be more plastic than others; for example, previous experience affected the amount of eye contact made during the unsolvable task much more than it affected the ability to read human gestures (Table 5). Finally, this type of research with dogs also has the potential to provide a roadmap for testing the development and evolution of cognitive architecture in a wider range of species.

Author Contributions

Hannah Salomons: Writing – review & editing, Writing – original draft, Visualization, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Morgan Ferrans:** Writing – review & editing, Project administration, Investigation, Data curation. **Candler Cusato:** Writing – review & editing, Project administration, Investigation, Data curation. **Kara Moore:** Writing – review & editing, Project administration, Methodology, Investigation, Data curation. **Vanessa Woods:** Writing – review & editing, Project administration. **Emily Bray:** Writing – review & editing, Supervision, Conceptualization. **Brenda Kennedy:** Writing – review & editing, Supervision. **Theadora Block:** Writing – review & editing, Project administration. **Laura Douglas:** Writing – review & editing, Project administration. **Ashton Roberts:** Project administration. **Margaret Gruen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Brian Hare:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Data Availability

The raw data for this study are available as Supplementary Material.

Declaration of Interest

The authors declare that they have no conflicts of interest.

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Supplementary Material

Supplementary material associated with this article is available at <https://doi.org/10.1016/j.anbehav.2025.123410>.

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